Jordan Mark **Barbone**

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Data platform and statistical programming leader focused on clinical trials analytics and automation

EXPERIENCE

Cogstate — New Haven, CT (Remote)

Principal, Data Platform

Sep 2026 - Present

- *I just started – give me some time*

Senior Data Scientist

Oct 2024 - Sep 2026

- Refactored legacy data ingestion processes into **Databricks** centered, **PySpark** coded scheduled jobs with ~70% reduction of time and ~50% reduction in cost; ingestions managing **REST API** responses, and **xlsx**/**csv** files in **SharePoint** with insert, update, and delete history preservation
- Developed configuration schemas and validation process centered around **git** to allow broader team input into processes while maintaining data quality
- Developed logging and alerting systems in **Databricks** and **SQL** alerts for potential data quality issues to streamline responses to outside team members
- Maintained and improved Statistical Monitoring program in **R** for outcome anomaly detection for Clinical Trials
- Developed **SQL** queries for Tableau dashboards
- Refactored Statistical Monitoring program in **R** reducing runtimes by ~50% and improving development scaling and unit test coverage
- Provided supervision and input for Data Scientist, Data Engineer, and Data Visualization resources

Data Scientist

Jun 2021 - Oct 2024

- Independently developed internal **R packages** to advance programming, visualization, normative data management, central monitoring, data access, and general utilities
- Designed and development **R shiny** applications for displaying central monitoring activity and performing interactive analyses
- Refactored Data Engineering **PySpark** code into a **python** package for data transformations within **Azure Databricks** and **DeltaTable** management
- Refactored legacy **SQL** code to seamlessly integrate **PySpark** functionalities and enforce best coding practices
- Designed and implemented an efficient process for managing and updating and tracking normative data, collaborating closely with Science and Product Management teams
- Authored comprehensive standards and protocols for **Jira** workflows and **git** management, ensuring consistency and efficiency
- Developed and managed a robust daily and weekly reporting system with **R tidyverse** scripts for the monitoring of a high-profile international Alzheimer's Disease trial
- Analyzed and interpreted requirements from Science and Data Management to drive strategic advancements within our Data Lakehouse
- Developed, reported, and presented analyses with **R tidyverse** scripts for custom data queries related to both ongoing and concluded clinical, providing actionable insights for rater training and central monitoring
- Developed a **python** data extraction, reformatting, and storage processes for academic research studies, greatly enhancing accessibility for end-users
- Contributed to the creation of poster materials for industry conferences and presentations, showcasing research, developments, and services

Madrigal Pharmaceuticals — Conshohocken, PA

Principal Data Analyst

Dec 2019 - Jun 2021

- Developed internal **R packages** with **tidyverse** features for the creation of daily metric and reports
- Aggregated and organized data sources in STDM, ADaM, TFLS, and pre-finalized formats; compiled source data from multiple vendor data bases and reports; reconciled differences between databases using **R tidyverse** suite
- Analyzed efficacy, safety, and exploratory data for: ongoing phase 3 studies in NASH and NAFLD; ad hoc analyses for phase 2; biomarker and patient reported outcomes with **R**
- Reviewed and reported on statistical testing in **R** such as: ANCOVA, LS-Means, ROC, Kruskal-Wallis, Wilcoxon rank sum, t-test, correlations (Pearson product, Spearman rho, Kendall's tau)
- Trained and onboarded data analyst

Signant Health — Wayne, PA

Clinical Data Scientist

Jan 2019 - Nov 2019

- Developed analytic **R** scripts using **tidyverse** features for clinical trial operational and data management monitoring.
- Created **R** scripts for processing clinical trial outcome data for risk-based statistical quality monitoring
- Provided analytic support for internal and external research projects, including conference poster preparation

Clinical Data Specialist

Jan 2018 - Dec 2018

- Analyzed central monitoring data for clinical trials using **R** scripts
- Provided analytic support for internal and external research projects, including conference poster preparation
- Reviewed audio data from ADAS-Cog, MMSE, and ADCS-ADL administrations; provided feedback on standard administration and scoring, and submitted score change reviews for clinical trial sites

Clinical Associate

May 2016 - Dec 2017

- Reviewed clinical trial raters' qualifications and recommended standardized training programs.
- Developed a training manual in **Microsoft Word** for the standardized administration of ADAS-Cog and MMSE within clinical trials, along with additional training presentations in **Microsoft PowerPoint**, ensuring alignment with company standards
- Reviewed audio data from ADAS-Cog, MMSE, and ADCS-ADL administrations; provided feedback on standard administration and scoring, and submitted score change reviews for clinical trial sites
- Developed new **Excel** trackers for central review tracking for individual central reviewers; compiled reports and presented information for central reviewer management

Clinical Assistant

Aug 2015 - May 2016

- Reviewed clinical trial raters' qualifications and recommended standardized training programs
- Scheduled re-training appointments for clinical trial raters in collaboration with central quality reviewers
- Created and maintained study trackers and teleconference schedules in **Excel**

Learning Assistance & Resource Center

West Chester University of Pennsylvania, West Chester, PA

Writing Tutor

Aug 2014 - May 2015

- **Tutored** students in one-on-one or small group sessions to enhance writing skills and promote independent, active learning
- Monitored the effectiveness of the Academic Development Program and tracked student progress by maintaining daily activity logs, using **Bloom's Taxonomy**, and preparing biweekly student reports and semester evaluations

EDUCATION

West Chester University of Pennsylvania
Master of Arts

West Chester, PA
Aug 2017 - May 2019

- General Psychology, Advisor: Geeta Shivde, Ph.D.
- Thesis: The effects of participant-selected background music on executive function task performance
- GPA: 4.00

West Chester University of Pennsylvania
Non-degree Graduate Student

West Chester, PA
Aug 2016 - Jun 2017

- Classes in Statistics, Research Methods, and Psychometrics
- GPA: 4.00

West Chester University of Pennsylvania
Bachelor of Arts

West Chester, PA
Aug 2011 - May 2015

- Major in Psychology, Minor in Biology
- Cumulative GPA: 3.470
- Major GPA: 3.834

SKILLS

R	statistical programming, package development, Shiny app development, tidyverse pipelines, Quarto/R Markdown reporting, and trial monitoring analytics
Python	PySpark data engineering, Databricks jobs, reusable package-based transformations, API and file ingestion workflows, and data automation
SQL	query development for analytics and dashboards, data quality checks, and alert-oriented monitoring workflows
Data Platforms	Databricks, Delta Lake patterns, CI/CD with GitHub Actions, configuration-driven processing, and operational logging/alerting
Clinical Research Analytics	clinical trial data review, centralized/risk-based monitoring, cognitive outcomes support, and rater-focused analytics and reporting
Collaboration & Workflow	Git-based collaboration, standards development, cross-functional data/science partnership, and reproducible analysis practices

SELECTED AWARDS

- **Cogstate Values Award** (Dec 2022) — Cogstate
  In recognition of outstanding dedication and contributions for furthering the Cogstate mission of improving brain health around the world.
- **Employee Recognition Award** (Nov 2021) — Cogstate
  Modeling one of Cogstate values through significant achievement: Lead with Science
- **WCUPA Psychology Service Award** (Apr 2015) — Psychology Department, West Chester University of Pennsylvania
  Dedicated service to the Psychology Department through positions in undergraduate club, a student-faculty committee, and the honor society
- **Psychology Research Achievement Award** (Apr 2015) — Psychology Department, West Chester University of Pennsylvania
  Involvement in research activities, poster presentations, and work on conducting and reporting on a meta-analysis for publication
- **CRLA Certified Tutor Level II** (Apr 2015) — Learning Assistance and Resource Center, West Chester University of Pennsylvania
  Successfully completed standardized CRLA training for tutors
- **Dean's List** (Dec 2014) — College of Arts and Sciences, West Chester University of Pennsylvania
  Completed 12 or more graded hours in an academic semester and achieve a semester GPA of 3.670 or better
- **CRLA Certified Tutor Level I** (Dec 2014) — Learning Assistance and Resource Center, West Chester University of Pennsylvania
  Successfully completed standardized CRLA training for tutors
- **Dean's List** (May 2013) — College of Arts and Sciences, West Chester University of Pennsylvania
  Completed 12 or more graded hours in an academic semester and achieve a semester GPA of 3.670 or better

COURSES

- *Data Scientist with R Track* — DataCamp
- *Statistician with R Track* — DataCamp
- *R Programmer Track* — DataCamp
- *Data Analyst with R Track* — DataCamp
- *R Programming* — DataCamp
- *Importing & Cleaning Data with R Track* — DataCamp
- *Machine Learning Fundamentals in R Track* — DataCamp
- *Statistical Fundamentals with R Track* — DataCamp
- *Statistical Fundamentals with R Track (Old Track)* — DataCamp
- *Python Programming Track* — DataCamp
- *SQL Fundamentals Track* — DataCamp
- *SAS Programming 1: Essentials* — SAS
- *SAS Programming for R Users* — SAS

TUTORING

- , Effective Writing (Aug 2014 - May 2015) — West Chester, PA
- , Psychology of Learning (Aug 2014 - May 2015) — West Chester, PA
- , Introduction to Biopsychology (Aug 2014 - Dec 2014) — West Chester, PA
- , Biopsychology of Motivation & Emotion (Aug 2014 - Dec 2014) — West Chester, PA
- , Behavioral Statistics (Aug 2014 - May 2015) — West Chester, PA
- , Research Methods in Psychology (Aug 2014 - May 2015) — West Chester, PA

EXTRACURRICULARS

- **Chapter Graduate Liaison**, Psi Chi Chapter, Psychology Department, West Chester University of Pennsylvania (Dec 2017 - May 2019) — West Chester, PA
- **Graduate Member**, Psi Chi Chapter, Psychology Department, West Chester University of Pennsylvania (Aug 2017 - Dec 2017) — West Chester, PA
- **Peer Mentor**, West Chester University of Pennsylvania (Jan 2015 - May 2015) — West Chester, PA
- **Chapter Secretary**, Psi Chi Chapter, Psychology Department, West Chester University of Pennsylvania (May 2014 - May 2015) — West Chester, PA
- **Club Publicity Chair**, West Chester University of Pennsylvania (May 2014 - May 2015) — West Chester, PA
- **Student Member**, Student Life Committee, Psychology Department, West Chester University of Pennsylvania (Sep 2013 - May 2014) — West Chester, PA
- **Undergraduate Member**, Psi Chi Chapter, Psychology Department, West Chester University of Pennsylvania (Mar 2013 - May 2014) — West Chester, PA
- **Undergraduate Member**, West Chester University of Pennsylvania (Aug 2012 - May 2014) — West Chester, PA

R PACKAGES

- **Barbone, J. M.** (2026). Cnd: Create and register conditions. https://jmbarbone.github.io/cnd/. (v 0.2.0)
- **Barbone, J. M.** (2026). Fuj: Functions and utilities for jordan. https://jmbarbone.github.io/fuj/. (v 0.1.0)
- **Barbone, J. M.**, & Garbuszus, J. M. (2026). openxlsx2: Read, write and edit xlsx files. https://janmarvin.github.io/openxlsx2/. (v 0.3.0)
- **Barbone, J. M.** (2025). Mark: Miscellaneous, analytic r kernels. https://CRAN.R-project.org/package=mark. (v 0.8.3)
- **Barbone, J. M.** (2023). Echo: Echo code evaluations. https://github.com/jmbarbone/echo. (v 1.30)
- **Barbone, J. M.** (2023). Scribe: Command argument parsing. https://jmbarbone.github.io/scribe/. (v 0.3.0)

PAPERS

Peer Reviewed Publications

- Younossi, Z. M., Stepanova, M., Taub, R. A., **Barbone, J. M.**, & Harrison, S. A. (2021). Hepatic fat reduction due to resmetirom in patients with nonalcoholic steatohepatitis is associated with improvement of quality of life. Clinical Gastroenterology and Hepatology. doi:10.1016/j.cgh.2021.07.039
- Solomon, T. M., **Barbone, J. M.**, Feaster, H. T., Miller, D. S., deBros, G. B., Murphy, C. A., & Michalczuk, D. (2019). Comparing the standard and electronic versions of the Alzheimer's disease assessment scale—cognitive subscale: A validation study. The Journal of Prevention of Alzheimer's disease, 6(4), 237–241. doi:10.14283/jpad.2019.27
- Irani, F., **Barbone, J. M.**, Beausoleil, J., & Gerald, L. (2017). Is asthma associated with cognitive impairments? A meta-analytic review. Journal of Clinical and Experimental Neuropsychology, 39(10), 965–978. doi:10.1080/13803395.2017.1288802

Master's Thesis

- **Barbone, J. M.** (2019). The effects of participant-selected background music on executive function task performance [Master's thesis, West Chester University of Pennsylvania; West Chester University of Pennsylvania]. doi:10.13140/rg.2.2.34109.41446

POSTERS

Cogstate - Cognitive outcomes

- Tronchin, G., Kingery, L., Leventhal, R., Bartolic, E., Wacker, S., **Barbone, J. M.**, & Wessels, A. M. (2024, March). Central monitoring of rater performnace and characteristics of efficacy and assessments in the TRAILBLAZER-ALZ 2 study. International Conference on Alzheimer's and Parkinson's Diseases (ADPD).
- **Barbone, J. M.**, Khurshid, K., Edgar, C. J., & Maruff, P. (2023). Equating the ADAS-Cog and MMSE to the Cogstate Brief Battery scores. Alzheimer's Association International Conference (AAIC), 19, e077060. doi:10.1002/alz.077060

- Edgar, C. J., Maruff, P., Harrison, J. E., **Barbone, J. M.**, Leventhal, R., & Alam, J. j. (2022). Validity and reliability of a composite cognitive outcome measure for clinical trials in dementia with lewy bodies. International Conference on Alzheimer's and Parkinson's Diseases (ADPD). https://www.cogstate.com/wp-content/uploads/2022/04/OO171-Validity-and-reliability-of-a-composite-cognitive-outcome-in-DLB%5Fd273-Read-Only.pdf
- Wacker, S., & **Barbone, J. M.** (2022). Rater perspectives on Applied Training of cognitive clinical outcome assessments, delivered by Neuropsychology experts. Alzheimer's Association International Conference (AAIC). https://www.cogstate.com/wp-content/uploads/2022/08/Rater-Perspectives-on-Applied-Training-of-Cognitive-Clinical-Outcome-Assessments.pdf
- Edgar, C. J., Amariglio, R., **Barbone, J. M.**, Chandler, J., Coons, S., Donohue, M., Lenderking, W., & Sperling, R. (2022, December). Preliminary evidence for reliability and validity of the Interpersonal Functioning and Daily Activities Questionnaire (IFDAQ) in the A4/LEARN pre-randomization sample. Clinical Trials on Alzheimer's Disease (CTAD).

Madrigal - Liver disease

- Harrison, S. A., Taub, R., **Barbone, J. M.**, Franc, J., & Karsdal, M. A. (2020, March). Resmetirom, a beta selective thyroid hormone receptor agonist, reduces net collagen III deposition in nonalcoholic Steatohepatitis. AASLD Emerging Topic Conference: Nuclear Receptors in Nonalcoholic Fatty Liver Disease.
- Harrison, S. A., Taub, R. A., Karsdal, M. A., Franc, J., Bashir, M., **Barbone, J. M.**, Neff, G., Gunn, N. T., & Moussa, S. (2020, November). Algorithm for predicting advanced NASH fibrosis on screening biopsy in resmetirom phase 3 MAESTRO-NASH clinical trial. AASLD Liver Meeting Digital Experience.
- Younossi, Z. M., Stepanova, M., Taub, R. A., **Barbone, J. M.**, Moussa, S., & Harrison, S. A. (2020, November). Improvement of health-related quality of life is associated with improvement of fat fraction by MRI-PDFF in patients with nonalcoholic steatohepatitis treated with resmetirom. AASLD Liver Meeting Digital Experience.

Signant Health (Bracket) - eCOA data quality

- Crittenden, K., Machizawa, S., Todd Feaster, H., **Barbone, J. M.**, Hong, B., Verma, P., & Zhou, W. (2020). Cross-cultural differences in PANSS item ratings: Comparisons of six geo-cultural regions. Schizophrenia International Research Society (SIRS) Conference, 46, S46–s46. doi:10.1093/schbul/sbaa031.103
- Roy, M., Brown, J., Hong, B., Benecke, R., Chen-Tackett, Z., Zhou, W., **Barbone, J. M.**, Feaster, H. T., & Sachs, G. (2020, September). eCOA prompted MADRS interview: Balancing through assessment and efficiency. ISCTM Annual Scientific Meeting.
- Feaster, H. T., **Barbone, J. M.**, Miller, D. S., & Solomon, T. M. (2019). Rater remediation on the ADAS-cog leads to longitudinal improvement in clinical trial data quality. Alzheimer's Association International Conference (AAIC), 15, P1377–p1377. doi:10.1016/j.jalz.2019.06.3923
- Karas, S. M., **Barbone, J. M.**, Solomon, T. M., Hong, B., & Feaster, H. T. (2019). Impact of data quality programs on MMSE inclusion criteria in Alzheimer's disease clinical research trials. Alzheimer's Association International Conference (AAIC), 15, P361–p361. doi:10.1016/j.jalz.2019.06.862
- Solomon, T. M., Feaster, H. T., Karas, S. M., Garcia-Valdecasas, M., **Barbone, J. M.**, DiGregorio, D. T., DeBonis, D., & Miller, D. S. (2019). The evolution of technology in data quality programs for Alzheimer's disease clinical trials: What have we learned and where are we going. Alzheimer's Association International Conference (AAIC), 15, P162–p162. doi:10.1016/j.jalz.2019.06.4332
- Feaster, H. T., **Barbone, J. M.**, Garcia-Valdecasas-Colell, M., & Solomon, T. M. (2018). Electronic ADAS-Cog (eADAS-Cog) data quality: How do countries compare? Alzheimer's & Dementia: Volume 14, Issue 7S_part_19, 14, P1069–p1069. doi:10.1016/j.jalz.2018.06.1373
- Karas, S. M., **Barbone, J. M.**, DeBonis, D., & Solomon, T. M. (2018). The use of statistical modeling to complement data quality programs. Alzheimer's Association International Conference (AAIC), 14, P295–p295. doi:10.1016/j.jalz.2018.06.066
- Solomon, T. M., **Barbone, J. M.**, Miller, D. S., & Feaster, H. T. (2018). Pilot validation of an Electronic Alzheimer's Disease Assessment Scale - Cognitive Subscale (eADAS-Cog). Alzheimer's Association International Conference (AAIC), 14, P680–p680. doi:10.1016/j.jalz.2018.06.719
- Solomon, T. M., Karas, S. M., **Barbone, J. M.**, DiGregorio, D. T., Miller, D. S., & Feaster, H. T. (2018). Analysis of the rates and types of errors on the enhanced eCOA version of the Alzheimer's Disease Assessment Scale - Cognitive subscale and Mini-mental State Examination used in dementia clinical trials. Alzheimer's Association International Conference (AAIC), 14, P296–p297. doi:10.1016/j.jalz.2018.06.070
- Seichepine, D. R., Solomon, T. M., Jacobs, E. B., **Barbone, J. M.**, DiGregorio, D. T., & Miller, D. S. (2018, February). Comparison between "flat" and "enhanced" eCOA of MMSE Attention and Calculation in clinical trials of Alzheimer's disease. ISCTM Annual Scientific Meeting. doi:10.13140/rg.2.2.16193.61289
- **Barbone, J. M.**, Solomon, T. M., Feaster, H. T., Garcia-Valdecasas-Colell, M., & Miller, D. S. (2018, October). MMSE screening data quality for Alzheimer's disease studies across countries. Clinical Trials on Alzheimer's Disease (CTAD). doi:10.13140/rg.2.2.27921.68966
- Garcia-Valdecasas-Colell, M., **Barbone, J. M.**, Solomon, T. M., Feaster, H. T., & Tott, N. (2018, October). Identifying impact of rater change on MMSE and CDR data in multi-national Alzheimer's disease clinical trials. Clinical Trials on Alzheimer's Disease (CTAD). doi:10.13140/rg.2.2.21210.80327
- Wessels, A. M., **Barbone, J. M.**, DiGregorio, D. T., Miller, D. S., Mullen, J. A., Sims, J. R., & Solomon, T. M. (2018, October). Lanabecestat-Rater performance and error characteristics of efficacy assessments in the DAYBREAK-ALZ study. Clinical Trials on Alzheimer's Disease (CTAD). https://bit.ly/32pl8au
- Butler, A., Feaster, H. T., Miller, D. S., & **Barbone, J. M.** (2017). Using iterative user experience design to improve electronic clinical outcomes assessment data quality. Value in Health, 20, A313–a313. https://bit.ly/3isANeP
- Feaster, H. T., Solomon, T. M., Abi-Saab, D., Vogt, A., **Barbone, J. M.**, Harrison, J., & Miller, D. S. (2017). The impact of electronic clinical outcome assessments (eCOA) on Alzheimer's disease clinical trial data quality. Alzheimer's Association International Conference (AAIC), 13, P935. doi:10.1016/j.jalz.2017.06.1828

- Enger, B., & **Barbone, J. M.** (2017, October). Distribution of performance and incorrect rantings in qualification video scoring. CNS Summit. doi:10.13140/rg.2.2.14096.46088
- Solomon, T. M., **Barbone, J. M.**, Karas, S. M., & Feaster, H. T. (2017, October). Longitudinal impact of audio review on data quality. Clinical Trials on Alzheimer's Disease (CTAD). https://bit.ly/2DYaFtd
- Solomon, T. M., Feaster, H. T., **Barbone, J. M.**, & Miller, D. S. (2017, November). Utilizing audio review to improve ADCS-ADL data quality. Clinical Trials on Alzheimer's Disease (CTAD). https://bit.ly/3mlKuOc

WCUPA - Asthma and music cognitive

- **Barbone, J. M.**, & Shivde, G. (2019, May). The effects of participant selected background music on executive task performance. Association for Psychological Science Annual Convention. doi:10.13140/rg.2.2.26125.79842
- **Barbone, J. M.** (2018, November). Trends in self-reported use of music while studying: Implications for research. West Chester University Research and Creative Activity Day. doi:10.13140/rg.2.2.14075.57124
- **Barbone, J. M.**, Saxena, S., & Irani, F. (2015, February). Neuropsychological performance in individuals with asthma: Two meta-analyses. Eastern Psychological Association Conference. https://bit.ly/3hu6J0i
- Mulligan, R. D., **Barbone, J. M.**, Saxena, S., Carey-Jr, D., Clappsy, J., & Irani, F. (2015, March). Asthma cognition data collection. West Chester University Psychology Research Day. https://bit.ly/2ZAD5kn
- **Barbone, J. M.**, Saxena, S., & Irani, F. (2015, July). Does asthma effect neuropsychological performance? A meta-analysis. American Psychological Association Convention.
- Irani, F., **Barbone, J. M.**, & Beausoleil, J. (2015, August). The impact of asthma on cognitive functioning. American Psychological Association Convention.